

Saddleback Dam Mitigation

Background

The Saddleback Flood Retarding Structure (FRS) is a 5-mile-long compacted earth-fill dam with a principal spillway and four irrigation outlets. The Flood Control District of Maricopa County, Arizona, identified the need to modify the embankment of the dam to mitigate distress observed in the form of numerous erosion holes and longitudinal cracks, along the crest of the dam.

Scope

Gannett Fleming performed numerous geotechnical investigations, developed mitigation alternatives, and prepared final design plans and specifications. We investigated three potential causes of the observed distress: settlement of collapsible foundation soil, incompatibility of the filter/drain with the overlying embankment soil, and settlement of the potentially loosely placed filter/drain. Our firm also performed failure modes and effects analyses to assess the mitigation alternatives for the central filter/drain.

In addition, our firm also prepared an economic assessment of flood damages in the 100-year storm event, both with and without the dam in place, based on FLO-2D modeling. The analysis included the assessment of flood damages to a downstream natural gas-powered combined-cycle electric generating plant. This analysis provided quantitative input on potential damages to use in the conceptual development of rehabilitation alternatives.

Accomplishments

Gannett Fleming investigated the cause of the distress on the Saddleback FRS and developed several mitigation measures and final design to rehabilitate the structure. Flood damage modeling and the economic impact assessment for the 100-year storm event were also conducted and factored into the final design and specifications.



The Saddleback FRS, prior to rehabilitation, was observed to have numerous erosion holes and longitudinal cracks, along the crest of the dam.

Client

Flood Control District of Maricopa County

Location

Maricopa County, Arizona

Firm's Fee

\$455,000